

3(a)	(10 – 5) / 5 × 100 ; 100 ;	2
3(b)	<u>restriction</u> , enzyme / endonuclease ; reverse transcriptase ;	2
3(c)	<i>any two from:</i> 1 small / low mass, so can enter, cells / <i>E. coli</i> (through membrane); 2 self-replicate in cell so multiply / make many copies of, gene ; 3 have restriction site(s) / can be cut by restriction enzymes, so new gene can, be added / join ; 4 have, marker / antibiotic resistance / fluorescence / reporter, <u>genes</u> , so , recombinants / transformed cells / cells that took up plasmid, can be recognised ; 5 have promoter so gene can be, expressed / transcribed ; 6 circular so , more stable / not damaged by host cell enzymes ;	2
3(d)	<i>any four from:</i> 1 prevents / less / lower, (named) disease / infection / pathogen transmission, risk ; 2 large / unlimited, supply OR mass production (from engineered bacteria) ; 3 cost of, purification / processing, lower ; 4 lower risk of, allergy / immune reaction / rejection / side effects ; 5 potential to, engineer / improve, recombinant proteins ;	4
3(e)(i)	fewer genes, expressed / have increased expression ;	1
3(e)(ii)	<i>any two from:</i> 1 transcription factors ; 2 (help / stop), binding / functioning, of RNA polymerase ; 3 ref. promoter ; 4 AVP ; light, is detected (by eyes) / causes changes melatonin enhancer / silencer, (DNA) sequences signalling, molecules / pathways	2

Question	Answer	Marks
3(e)(iii)	<i>any two from:</i> 1 <i>idea of</i> compare with known, genes / sequences / genomes, in database ; 2 search / analysis, programme / software / algorithm ; 3 identify, role of protein / (named) health effects ;	2

Question	Answer	Marks
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